

ID5003W - DA5402W: MLOps Lab



Coordinators:

Prof. Sridharakumar Narasimhan and Prof. Nirav Bhatt

Course Instructors:

Dr. Alka Bhushan, Dr. Priyanka Naik, Dr. Aishwariya Chakraborty

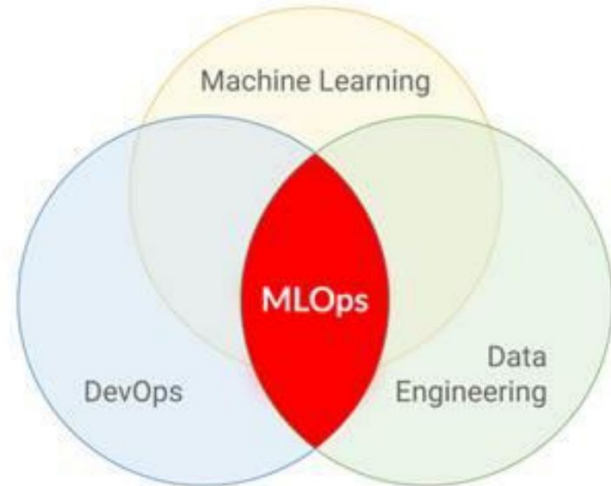
Head Teaching Assistants:

Jashaswaimalya Acharjee, Shuvrajeet

MLOps : Machine Learning Operations

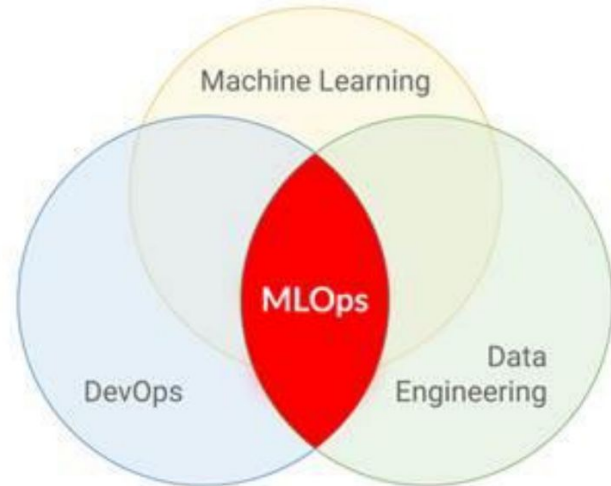
MLOps : Machine Learning Operations

- An engineering discipline that brings ML Models into production reliably and efficiently.
- Manages end-to-end ML lifecycle : data -> training -> deployment -> monitoring



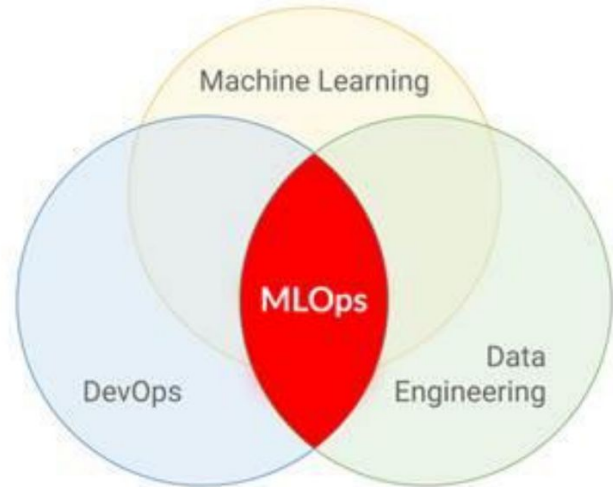
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- Tasks:



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- Tasks:
 - Experiment Tracking
 - Model versions, evaluation and accuracy metrics, hyperparameters, dataset versions, training configurations, execution logs
 - Model Deployment
 - APIs, web services, edge devices, cloud platforms
 - Model monitoring
 - Prediction accuracy, response latency, resource utilization, failure rates, data quality, drift detection
 - Model Retraining
 - Scheduled, trigger based, continuous



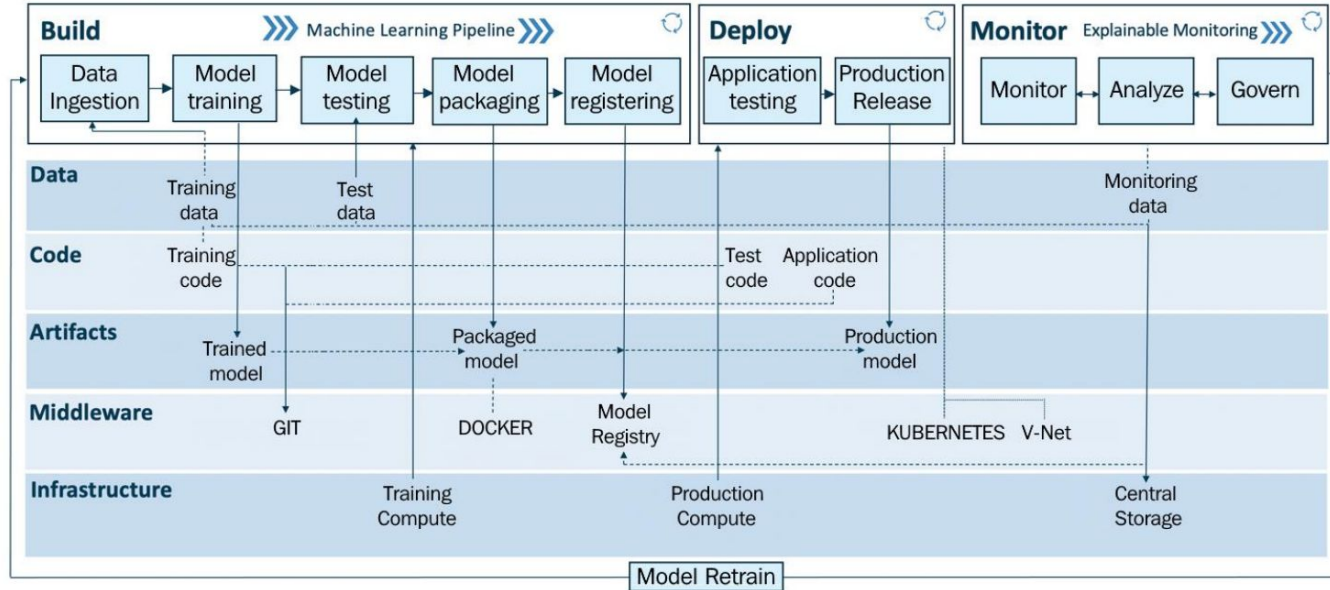
MLOps Benefits

- Manages complex ML systems and large-scale data pipelines
- Enables scalable and reliable model deployment
- Supports automation through CI/CD/CT pipelines
- Improves maintainability and adaptability
- Enables monitoring, drift detection, and continuous improvement
- Enhances reproducibility and collaboration

ML Workflow

Segmented into two modules : (i) MLOps pipeline (ii) Driver

MLOps Workflow



Tools and Technologies

- **Apache Airflow** : automate and schedule pipeline tasks
- **Apache Kafka** : handles real-time streaming data
- **Apache Spark** : process and analyze large scale data, distributed machine learning
- **MLFlow** : manage experiment tracking, model lifecycle and deployment
- **AutoML** : automate model selection, training and tuning
- **Git & DVC** : version control for data, models and ML pipelines
- **Docker & Kubernetes**: containerization and orchestration of scalable applications
- **Prometheus & Kibana** : data visualization, monitoring of logs and metrics

Objectives and Learning outcomes of the Lab

- The objective of this laboratory is to provide hands-on experience in building and deploying scalable AI solutions for industrial applications
- **Learning outcomes:** At the end of this lab, students should be able to
 - Use Apache Spark distributed platform for data preprocessing, feature engineering, and ML modeling.
 - Implement ML pipelines using Apache Airflow
 - Manage reproducibility and version control using MLflow and DVC.
 - Explore scalable architectures, including GPU-based and cloud systems.
 - Visualize data, system metrics and model performance using Prometheus and Kibana.
 - Develop end-to-end scalable AI solutions tailored to industrial environments.

Books and Reference Materials

- Emmanuel Raj, “Engineering MLOps”, Packt publication, 2021
- Chip Huyen, “Machine Learning Systems Design”, O’Reilly Media, 2022
- Jules S. Damji, Brooke Wenig, Tathagata Das, Denny Lee. “*Learning Spark: Lightning-Fast Data Analytics (2nd Edition)*”, O'Reilly Media, 2020.
- Adi Polak, “*Scaling Machine Learning with Spark Distributed ML with MLlib, TensorFlow, and PyTorch*”, O'Reilly Media, 2023.

Course Content

Module 1: Introduction to MLOps

- MLOps vs Traditional Software Development
- Key Challenges in Production ML
- MLOps Components, Workflow & Tooling Landscape

Module 4: Versioning and Collaboration

- Git, GitHub/GitLab
- Git LFS, DVC for Data & Model Versioning

Module 2: Data Engineering

- Data Pipelines (ETL, Batch & Streaming)
- Tools: Apache Airflow, Kafka, Spark, Beam, Flume
- Data Quality, Governance, Cleaning & Imbalance Handling

Module 5: Deployment and Serving

- Docker, Cloud, Serverless (AWS Lambda)
- TensorFlow Serving, FastAPI
- Edge/On-Prem Deployment, TensorRT Optimization

Module 6: Monitoring and Optimization

- Model Monitoring & Drift Detection
- Logging (Kibana), Metrics (Prometheus)
- Scaling & Performance Optimization

Module 3: Model Development

- ML Frameworks & Feature Engineering
- MLflow, Ray (Tracking & Distributed Training)
- Streaming ML, MLlib (Regression/Clustering)
- Deep Learning, Transfer Learning, AutoML

Module 7 : CI/CD and Governance

- CI/CD Pipelines (Jenkins, GitLab CI/CD, Cloud Tools)
- TFX Pipelines
- Security, Explainability & Responsible AI

Datasets for Module 1-3

- Intel Image Classification dataset :
<https://www.kaggle.com/datasets/puneet6060/intel-image-classification/data>
- NYC Taxi Trip Records:
https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page?utm_source=chatgpt.com

Course Information

- Course Instructors:
 - Dr. Alka Bhushan, Dr. Priyanka Naik, Dr. Aishwariya Chakraborty
- Course TAs:
 - Lead TAs : Jashaswaimalya Acharjee (tools/software related issues), Shuvrajeet (assignment and quiz related issues)
 - TAs :
- Course Timings
 - **Thursday 5.30 - 7.00 pm**
 - **Friday 5.30 - 7.00 pm**
- In each week :
 - One demo + teaching class
 - One practice / doubt session
- Total Weeks : 13

Grading Policy

- Assignments
 - 3-5 lab assignments
 - 2 Week time to submit assignments
- Quizzes, and End Term Exams
 - 2 quizzes
- End Term Project
 - Group assignment
 - Submit problem and data
 - Submission will be 2 weeks before end term
 - Viva will be conducted
- Grading Weightage: 35% assignments, 20% quizzes, 5% class quizzes, 40% end term project
- Copying cases : penalty by the institute and will be informed to the company

A Simple Scripts based Automation Pipeline